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Hardware

Nano Matter



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Getting Started with Nano Matter Display

Getting Started with Nano Matter Display

This tutorial teaches you the basics about the Nano Matter Display, leveraging the Arduino environment to develop E-Ink + Matter applications.

Author · Christopher Méndez · Last revision · 02/17/2025

Overview

The [Nano Matter Display](#) is a development kit for the Nano Matter that features an extension board from [Pervasive Displays](#). It includes a **2.9 inches (384x168 resolution) E-ink monochrome high-definition display**, a temperature and humidity sensor, a 3-axis accelerometer and an addressable RGB LED.



The Nano Matter Display has been created to be used along the Arduino Nano Matter to extend its capabilities and allow the user to create interesting and useful applications around the E-ink display technology.

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Goals

Learn how to use the Nano Matter Display and all its features

Learn how to develop a Matter application using the Nano Matter Display with the Arduino Nano Matter

Required Hardware and Software



Nano Matter Display
Hardware

Hardware Requirements

[Arduino Nano Matter](#) (x1)

Nano Matter Display extension board (x1)

[USB Type-C® Cable](#) (x1)

Software Requirements

[Arduino IDE](#) or [Arduino Cloud Editor](#)

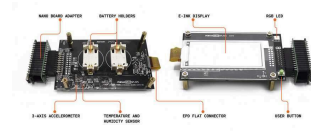
[Silicon Labs boards package](#)
(to support the Nano Matter within the Arduino IDE)

Pervasive Displays EPDK-Matter Official Documentation

To learn more about the EPDK-Matter Development Kit read the Pervasive Displays [documentation](#).

Nano Matter Display

Main Features



Nano Matter Hardware
Overview

The **Nano Matter Display** include the following features:

E-ink display: a 2.9 inches (384x168) E-ink monochrome high-definition display with a wide temperature working range and fast update.

RGB LED: a WS2813C addressable RGB LED for visual status feedback.

3-axis accelerometer: a MEMS digital output motion sensor (LIS2DH12).

Temperature and humidity sensor: a HDC2080 Low-Power humidity and temperature digital sensor.

Flexible power options: can be powered by a pair of CR2450 3V batteries or through the USB connector of the Arduino Nano.

Nano Matter Display Kit Assembly

In the **Nano Matter Display kit** you will find the following parts:

E-ink display (2.9" 384x168).

**Pervasive Displays EPD
Extension Kit Gen 4 (EXT4).**

Pervasive Displays EXT4-to-Nano adapter.

Plastic washers (x4).

Acrylic sheet to protect the display (x1).

Follow the next animation for the assembly steps:



To connect the E-ink display, pull up gently the black lever on the connector and slide in the display flat cable with the copper contacts facing down, it will go in like 2mm. Finally, push down the connector lever so it latches the cable.

Align the E-ink display with the white silkscreen outlines in the other side of the PCB. Use a thin double-sided tape to fix it (not included).

The EXT4-to-Nano adapter must be connected so its **11-10** white mark is aligned with the **11-10** breakout pins of the Matter Display board.



Adapter connection

Configuring the Development Environment

To start using the Nano Matter Display we first need to install some basic dependencies for the Arduino

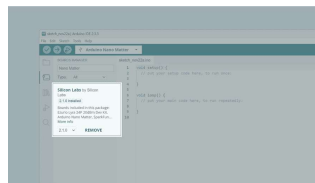
Arduino IDE Download

We are going to use the Arduino IDE to develop around the Nano Matter Display. Follow this guide to learn how to install the IDE if you have not done so.

Arduino IDE Installation Guide

Board Core and Library Installation

The **Silicon Labs** core contains the libraries and examples you need to work with the board's components, such as its Matter, Bluetooth® Low Energy, and I/Os. To install the Nano Matter core, navigate to **Tools > Board > Boards Manager** or click the Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for `Nano Matter` and install the latest `Silicon Labs` core version.



Installing the Silicon Labs core in the Arduino IDE

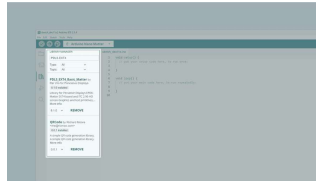
Download the following libraries:

`PDLS_EXT4_Basic_Matter.h`:

enables support for Pervasive Displays extension board to control the E-ink display and every other features. You can install it from the Library Manager on the Arduino IDE or from its [repository](#).

`qrcode.h`: allows you to create QR codes to then be displayed on the E-ink display. You can

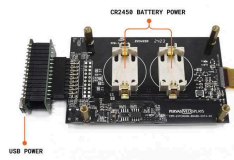
Manager on the Arduino IDE or from its [repository](#).



Libraries installation using the Library Manager

Follow these guides to learn more about library installation process ([Library Manager](#), [.zip install](#))

Powering the Kit



Matter Display power options

You have several options to power your EXT4 EPD development board that are listed below:

USB power: you can power the board using a USB-C cable connected to the Arduino Nano Matter.

Battery power: you can use two CR2450 batteries to power your kit. Recommended for remote deployments without access to wired power.

To avoid damaging your board only use one power source at a time. If USB power is used make sure not to have batteries installed.

Matter RGB Lightbulb

In order to quickly understand the product capabilities we are going to jump straight to a hands-on section based on the [RGB Lightbulb example](#) showcased in the [Nano Matter User Manual](#).



RGB Lightbulb Overview

As this example uses the Matter network you will need a **Thread**



Border Router to replicate it.

See the supported ones on the [Matter section of the Nano Matter User Manual](#).

The following example code is based on the Matter library built-in example called **nano_matter_lightbulb_color** and the **Pervasive Displays library API**, copy and paste it from the code snippet below:



```

1  #include <Matter.h>
2  #include <MatterLightbulb.h>
3  #include "PDLS_EXT4_Basic.h"
4  #include "hV_Configuration.h"
5  #include "ezWS2812gpio.h"
6  #include "qrcode.h"
7
8  uint8_t r, g, b;
9
10 MatterColorLightbulb matter;
11
12 const pins_t nano_matter_pins = {
13
14   ezWS2812gpio myRGB(1, r, g, b);
15
16   Screen_EPD_EXT4_Fast EInkDisplay;
17
18   #define MATTER_EXAMPLE_BUTTON_PIN 5
19
20   static uint8_t countFlashes = 0;
21   const uint8_t FAST_BEFCOUNT = 10;
22
23   volatile bool button_pressed = false;
24
25   void setup() {
26       Serial.begin(115200);
27
28       // Nano Matter device

```

Some of the code main functions to pay attention to are:

The `displayCommissioning()` function displays the commissioning steps on the screen, letting you know how to device status and showing the QR code.

The `displayQR()` function converts the Nano Matter onboarding QR payload on an image to be shown in the E-ink display.

The `loop()` function continuously checks if the user button was pressed and updates the LED color with the data received from the Matter controller.

The `setup()` function initiates the Matter service, E-ink display and other peripherals.

Using the Nano Matter Display will allow you to enjoy the following new features:

Commissioning QR Code displayed on the E-Ink screen so we will be able to scan it with your phone.



QR code on commissioning process

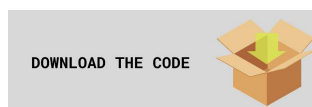
Real time data monitoring on the E-ink display showing the lightbulb state and colors.



Real time RGB color data

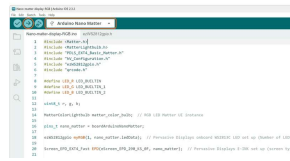
Upload the Example Sketch

You can copy the code from the previous section and paste it on the Arduino IDE or download it from this [link](#) or by clicking on the image below:



[Download the complete code here](#)

In the Arduino IDE select the **Arduino Nano Matter** inside the *Silicon Labs* board package and make sure the **Protocol Stack** is set to *Matter*.



RGB example code upload

Commissioning the Matter RGB Lightbulb and Final Test

Once the code is uploaded, the display will show the commissioning steps alongside the QR code.

See the following video that shows the whole process with the Amazon Alexa app.



Make sure to use the right assistant app respectively to your Thread Border Router brand. Here are the suggestions for [Google Home](#), [Amazon Alexa](#) and [Apple Home](#)

Matter Weather Station

As another hands-on application, we are going to create a **weather station** using our Matter Display that measures `temperature` and `humidity` while showing the readings on the screen and sharing the measurements with our



Weather Station Overview

As this example uses the Matter network you will need a **Thread**



Border Router to replicate it.

See the supported ones on the

[Matter section of the Nano](#)

[Matter User Manual](#).

Download the following library using the Arduino IDE Library Manager:

`HDC2080.h` : includes the support for the temperature and humidity sensor. You can install it from the Library Manager on the Arduino IDE or from its [repository](#).

The following code is based on the **Pervasive Displays** example included in the library called **EXT4_Matter_Weather**, copy and paste it from the code snippet below:



```

1  #include <Matter.h>
2  #include <MatterTemperature.h>
3  #include <MatterHumidity.h>
4  #include "PDLS_EXT4_Base.h"
5  #include "hV_HAL_Peripherals.h"
6  #include "hV_Configuration.h"
7  #include "Wire.h"
8  #include "qrcode.h"
9  #include <HDC2080.h>
10
11 MatterTemperature myMatterTemp;
12 MatterHumidity myMatterHumid;
13
14 pins_t nano_matter = {
15     .pin_m0 = 2,
16     .pin_m1 = 3,
17     .pin_m2 = 4,
18     .pin_m3 = 5,
19     .pin_m4 = 6,
20     .pin_m5 = 7,
21     .pin_m6 = 8,
22     .pin_m7 = 9,
23     .pin_m8 = 10,
24     .pin_m9 = 11,
25     .pin_m10 = 12,
26     .pin_m11 = 13,
27     .pin_m12 = 14,
28     .pin_m13 = 15,
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739    .pin_m724 = 726,
740    .pin_m725 = 727,
741    .pin_m726 = 728,
742    .pin_m727 = 729,
743    .pin_m728 = 730,
744    .pin_m729 = 731,
745    .pin_m730 = 732,
746    .pin_m731 = 733,
747    .pin_m732 = 734,
748    .
```


on an interesting place at home to monitor.



Weather Station
Deployment at home

Nano Matter Display Sensors and Actuators

In the following sections we are going to highlight some of the **Nano Matter Display** main features and learn how to use them.

E-Ink Display

The API basic functions to control the **E-ink display** are shown below.

Define the screen object using the Pervasive Displays class and initialize it:

```
1 Screen_EPD_EXT4_Fast <screen-name>  
2  
3 <screen-name>.begin();
```

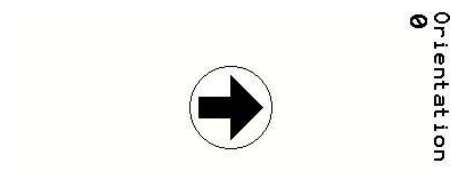
Replace the `<screen-name>` with a custom designation of your display for further usage. For this guide we are going to use `EPD`.

Screen Orientation

- 0: default
- 1: right rotated
- 2: reverse default
- 3: left rotated
- 6: ORIENTATION_PORTRAIT
- 7: ORIENTATION_LANDSCAPE

Define the screen orientation using the following function:

```
1 EPD.setOrientation(7);
```



Test the library example called [Common_Orientation](#)

Text

To display text on the E-ink screen you need to define different screen parameters such as **font**, **orientation**, the **string** to be shown and its **color**.

Fonts available:

Font_Terminal6x8

Font_Terminal8x12

Font_Terminal12x16

Font_Terminal16x24

Define the font using the following function:

```
1 EPD.selectFont(Font_Termin
```

Text writing:

Display text using the `gText` function, pass the coordinates and string:

```
1 EPD.gText(<x_coordinate>,
```

For example:

```
1 EPD.gText(0, 0, "Hello Wor
2 EPD.flush(); // update the
```

Colors:

The `gText` function also admits color definition for the text and the screen background. These are the available colors for this

Monochrome display:

`myColours.black` : default

`myColours.white` : use when background color is not *white*

`myColours.gray` : simulates gray using black and white dots (not gray scale).

For example:

```
1 EPD.gText(0, 0, "Hello Wor
2 EPD.flush(); // update the
```

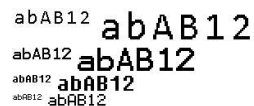
To print inverted color text you must clear the screen in black, then show white text using the following method:


```
1 EPD.clear(myColours.black)
2 EPD.gText(0, 0, "Hello Wor
3 EPD.flush(); // update the
```

Text Size:

You can choose between different fonts to achieve different text size or use the `gTextLarge` to make the text shown bigger.

```
1 EPD.gTextLarge(0, 0, "Hell
2 EPD.flush(); // update the
```



abAB12 abAB12
abAB12 abAB12
abAB12 abAB12
abAB12 abAB12

Text display



Test the library example called
[Common_Text](#)

Graphics

The E-ink display API includes some predefined functions for basic geometrical shapes, here you find some:

```
EPD.line(<x1>, <y1>, <x2>,  
<y2>, <color>)
```

```
EPD.circle(<x0>, <y0>,  
<radius>, <color>)
```

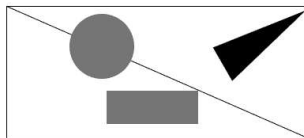
```
EPD.triangle(<x1>, <y1>,  
<x2>, <y2>, <x3>, <y3>,  
<color>)
```

```
EPD.rectangle(<x1>, <y1>,  
<x2>, <y2>, <color>)
```

```
EPD.point(<x1>, <y1>,  
<color>)
```

You can make them opaque or a wire frame using the `setPenSolid()` function as follows:

```
1 EPD.setPenSolid(true); //  
2  
3 EPD.setPenSolid(false); //
```



Forms display

Follow the [Pervasive Displays](#)  [documentation](#) for advanced graphics.

RGB LED

The Nano Matter Display features an addressable WS2813C RGB LED that can be easily used for visual status feedback.

The LED object is configured using the `ezWS2812gpio` class as follows:

```
1 const pins_t myBoard = bo  
2 ezWS2812gpio myRGB(1, myBo
```

To control the LED color use the `set_all` function as follows:

```
1 myRGB.set_all(<R>, <G>, <B>
```



LED demo



Test the library example called
[EXT4_WS2813C](#)

3-Axis Accelerometer

The Nano Matter Display features a LIS2DH12 3-axis accelerometer that can be easily used with open-source libraries as the `SparkFun LIS2DH12`, install it by using the IDE Library Manager.

The following example code adapts the screen orientation to the physical board rotation just as your smartphone does.



```
1  #include <Wire.h>
2
3  // EPD Screen Library
4  #include "PDLS_EXT4_Bas
5  #include "hV_Configura
6
7  // Accelerometer support
8  #include "SparkFun_LIS2
9
10 SPARKFUN_LIS2DH12 acce]
11
12 // PDLS
13 pins_t myBoard = board/
14
15 Screen_EPD_EXT4_Fast EF
16
17 uint8_t orientation = 0;
18 uint8_t oldOrientation
19 int aX, aY, aZ;
20
21 void setup() {
22   Serial.begin(115200);
23   delay(500);
24   Serial.println();
25
26   // Start
27   EPD.begin();
28   EPD.regenerate(); //
```

If you rotate the Nano Matter Display you will notice how the screen orientation will change making the arrow always to point upward.



Temperature and Humidity Sensor

humidity sensor that can be easily used with open-source libraries as the `Lime Labs HDC2080`, install it by using the IDE Library Manager.

```
1 #include "PDL5_EXT4_Bas
2 #include "hV_HAL_Peripl
3 #include "hV_Configurai
4 #include "Wire.h"
5 #include <HDC2080.h>
6
7 pins_t nano_matter = bc
8
9 Screen_EPD_EXT4_Fast EF
10
11 #define MATTER_EXAMPLE_
12
13 #define HDC_ADDR 0x40
14 HDC2080 sensor(HDC_ADDF
15
16 unsigned long previousM
17 const long interval = 5
18
19 struct measure_s {
20     float value;
21     float oldValue = 999.
22     float minimum = 999.9
23     float maximum = -999.
24 };
25
26 measure_s temperature;
27 measure_s humidity;
28
```

After uploading the example sketch you will be able to monitor the environment temperature and relative humidity alongside their historical minimum and maximums.



Temperature and Humidity

You can buy the Nano Matter Display Kit directly from the [Pervasive Displays](#) official store referenced below:

[Nano Matter Display Kit](#)

You can also buy some kit main components separately.

[2.9" E-ink display](#)

[EPD Extension Kit Gen 4 \(EXT4\)](#)

Conclusion

In this tutorial you learned how to use the Nano Matter Display expansion kit, leveraging all its features like the E-ink screen to display high-contrast graphics with a low power consumption on 2.9" 384x168 resolution. Features like the built-in RGB LED, the 3-axis accelerometer and the temperature and humidity sensor were explained.

Thanks to the E-ink display, your IoT-oriented products can have a better and more natural user experience while maintaining a low power consumption. Allowing you to set up and commission your solution by scanning the QR code and showing real-time sensor data on the screen, being a game-changer feature for your IoT solutions.

Next Steps

Extend your knowledge with E-ink displays following the [Pervasive Displays documentation](#).

Try all the examples included in

Start creating your own graphics to display custom data on the screen.

Suggest changes

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